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In-silico study of the interactions between acylated glucagon like-peptide-1 analogues and the native receptor

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ABSTRACT

We have performed a series of multiple molecular dynamics (MD) simulations of glucagon-like peptide-1 (GLP-1) and acylated GLP-1 analogues in complex with the endogenous receptor (GLP-1R) to obtain a molecular understanding of how fatty acid (FA) chain structure, acylation position on the peptide, and presence of a linker affect the binding. MD simulations were analysed to extract heat-maps of receptor–peptide interaction patterns and to determine the free energy of binding using the molecular mechanics Poisson-Boltzmann surface area (MM-PBSA) approach. The extracted free energies from MM-PBSA calculations are in qualitative agreement with experimentally determined potencies. Furthermore, the interaction patterns seen in the receptor–GLP-1 complex simulations resemble previously reported binding interactions validating the simulations. Analysing the receptor–GLP-1 analogue complex simulations, we found that the major differences between the systems stem from FA interactions and positioning of acylation in the peptide. Hydrophobic interactions between the FA chain and a hydrophobic patch on the extracellular domain contribute significantly to the binding affinity. Acylation on Lys26 resulted in noticeably more interactions between the FA chain and the extracellular domain hydrophobic patch than found for acylation on Lys34 and Lys38, respectively. The presence of a charged linker between the peptide and FA chain can potentially stabilise the complex by forming hydrogen bonds to arginine residues in the linker region between the extracellular domain and the transmembrane domain. A molecular understanding of the fatty acid structure and its effect on binding provides important insights into designing acylated agonists for GLP-1R.

Abbreviations: β 1- β 2: β -strand 1 to β -strand 2 turn; DPP IV: Dipeptidyl peptidase 4; ECD: Extracellular domain; ECDL2: Extracellular domain loop 2; ECLx: Extracellular loop x; E-TM: Extracellular domain to transmembrane domain linker; FA: Fatty acid; GLP-1: Glucagon-like peptide-1; GLP-1R: Glucagon-like peptide-1 receptor; GPCR: Glucagon protein-coupled receptor; HSA: Human serum albumin; ICLx: Intracellular loop x; MD: Molecular dynamics; MM-PBSA: Molecular mechanics Poisson-Boltzmann surface area; NPT: Constant number of particles, pressure, and temperature; PME: Particle-mesh-Ewald; RMSD: Root mean square deviation; SASA: Solvent accessible surface area; T2DM: Type 2 diabetes mellitus; TMD: Transmembrane domain; TMx: Transmembrane α -helix x

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1. Introduction

Lifestyle diseases such as cardiovascular diseases, diabetes, dementia, overweight, and obesity have significant health and economic costs (Geneva: World Health Organization, 2018a, 2018b; Sun et al., 2022). With regard to diabetes, type 2 diabetes mellitus (T2DM) comprises the majority of people with diabetes worldwide, and T2DM is partly caused by excess body weight and physical inactivity (Seidu et al., 2021; Sun et al., 2022). Different therapeutic treatments exist for T2DM, one such treatment is based on glucagon-like peptide-1 (GLP-1) which is involved in glucose-dependent insulin regulation (Drucker, 2021; Knudsen & Lau, 2019; Lund et al.,

1981). In addition, GLP-1 plays a role in suppressing gastric emptying and appetite, both features relevant to bodyweight regulation (Flint et al., 1998; Nauck et al., 1997; Read et al., 1994). Furthermore, studies have tied GLP-1 to neuroprotective properties, which could be of great importance in the treatment of dementia (Edison et al., 2021; Nørgaard et al., 2022; Zhao et al., 2015), and to reduce the risk of developing cardiovascular diseases (Drucker, 2016; Pahud de Mortanges et al., 2022; Ussher & Drucker, 2014). These highly attractive properties of GLP-1 are a result of the peptide's role in many different hormonal and signalling pathways where GLP-1 binds to the GLP-1 receptor (GLP-1R), which is present in many different tissues, e.g. the intestines, brain, heart, blood

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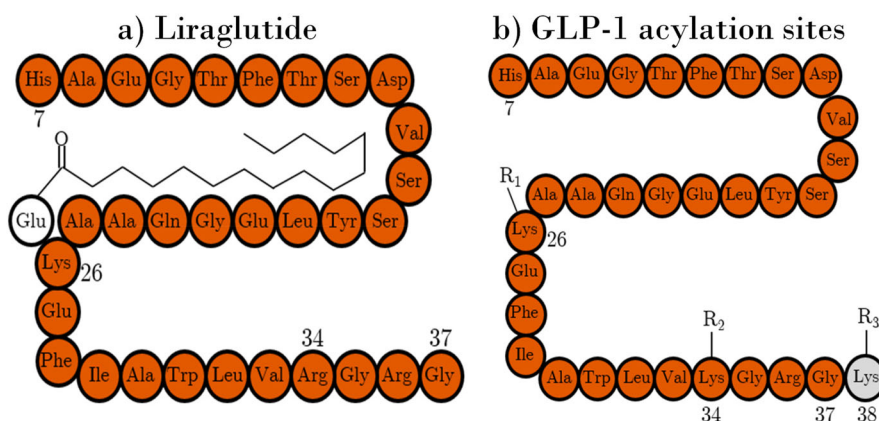


Figure 1. a) The sequence of liraglutide, including the attached fatty acid chain and Lys34*Arg mutation. b) The acylation sites of GLP-1 (Knudsen et al., 2000). Orange residues represent the base amino acid sequence corresponding to GLP-1, and the grey residue is an addition to certain analogues. R1, R2, and R3 indicate the position of the attached FA chains, either with or without linkers.

vessels, and pancreas (Alvarez et al., 2005; Deganutti et al., 2022; Graaf et al., 2016).

GLP-1 consists of either 30 or 31 amino acid residues (Holst et al., 1987; Orskov et al., 1989). The shorter version, GLP-1(7*-36*)-NH₂ ('*' notes peptide numbering) is slightly more abundant and stable in systemic circulation than the longer version, GLP-1(7*-37*), but both bind equally well to GLP-1R (Orskov et al., 1994; Wettergren et al., 1998). However, one particular challenge of using native GLP-1 is its short half-life in the bloodstream of approximately 2 min which stems from rapid degradation via dipeptidyl peptidase 4 (DPP IV), which cleaves the peptide between Ala8* and Glu9*, leaving the inactive GLP-1(9*-36*/37*) metabolite (Deacon et al., 1995). Additionally, GLP-1 is also cleared via the kidneys (Ruiz-Grande et al., 1993). To extend the half-life of GLP-1, different peptide analogues have been designed (Deacon, 2004; Drucker & Nauck, 2006; Lau et al., 2015) using strategies that e.g. include: *i*) acylation of GLP-1 with fatty acid (FA) chains, *ii*) covalent binding to human serum albumin (HSA), *iii*) attachment of Fc fragments of immunoglobulin, and *iv*) backbone elongation with a Lys tag in the C-terminal end (Ahrén, 2011; Bukrinski et al., 2017; Graaf et al., 2016).

This study will focus on GLP-1 analogues from group *i*). From this category, liraglutide, a Novo Nordisk drug sold under the name Victoza®, is worth mentioning. It was the first acylated GLP-1 analogue approved for T2DM treatment (Graaf et al., 2016; Knudsen et al., 2000) and later approved for obesity treatment. Liraglutide has been shown to control insulin secretion in a homeostatic fashion, which in turn, makes it possible to use the peptide for the treatment of other disorders in non-diabetic patients (Drucker et al., 2010). Liraglutide was also shown to have neuroprotective properties (Briyal et al., 2014; McClean & Hölscher, 2014; Talbot & Wang, 2014), as well as preventing cardiovascular diseases (Nauck, 2008). Liraglutide is a GLP-1(7*-37*) analogue that has a 16 carbon atom FA chain attached to Lys26* via an L-γ-Glu linker. Lys34* has been mutated to Arg34* to ensure selective acylation at Lys26* (Knudsen et al., 2000). The schematic structure of liraglutide is shown in Figure 1a.

For liraglutide and other GLP-1 acylated analogues, the attached FA chain can interact with HSA or form oligomers

induced by hydrophobic interactions, both preventing DPP IV degradation and renal clearance (Frederiksen et al., 2015; Knudsen et al., 2000; Peters, 1985). The result is a half-life extension to around 11 to 15 h (Knudsen et al., 2000).

In this study, several liraglutide-like GLP-1 analogues, including GLP-1(7*-36*)NH₂ and liraglutide, were investigated to study the effect of FA structure, FA length, acylation site, and linker on the binding of the peptide to the receptor. The selection was based on the study by Knudsen et al. (2000), who performed an extensive screening of different acylation strategies. Here, several candidates were selected to uncover the effects on the potency of different acylation schemes (Knudsen, 2010; Knudsen et al., 2000; Lau et al., 2015). The authors showed that potency (compared to that of native GLP-1) tends to increase with FA chain length up to 16 C-atoms where, for longer FA chains, potency decreases. The presence of a linker also showed an increase in potency. It was also shown that the addition of free acid in the FA chain ω tail decreases potency. In addition, acylation at the N-terminus of GLP-1 decreased potency, whereas acylation at the C-terminal end of the peptide did not affect potency (Knudsen, 2010). Here, a smaller selection of peptides was chosen from the study of Knudsen et al. (2000) to represent peptides with *i*) different acylation sites (shown in Figure 1b, Lys26*, Lys34*, and Lys38*), *ii*) different FA chain lengths (from 8-16 C-atoms), *iii*) the presence or absence of a linker, and *iv*), the presence or absence of a free acid group in the ω tail of the FA chain. Recently, a series of cryo-EM structures of the GLP-1R-peptide complex was deposited in the PDB (Deganutti et al., 2022; Zhang et al., 2017, 2021; Zhao et al., 2022). The cryo-EM structure of the GLP-1R-semaglutide complex has shown that the acylated side chain at Lys26* can either interact with TMD or the hydrophobic patch of the ECD (Zhang et al., 2021). However, the structure of receptor-bound liraglutide is still not solved and the effect of acylation on other sites such as Lys34* or Lys38* is not investigated in molecular details. *In silico* based approach provides an alternative option to study a complex behaviour of proteins to extend the knowledge that are determined from the experiments (Bottaro & Lindorff-Larsen, 2018). Molecular dynamics (MD) simulations have been previously used to study oligomerization process of liraglutide (Frederiksen et al., 2015), GLP-1 induced receptor activation

Table 1. The peptides are investigated with regard to the binding. 'Cxx' indicates the number of C atoms in the FA chain. 'L' indicates the presence of L- γ -Glu linker. 'A' indicates the presence of an ω acid group. 'da' indicates de-amino-His7*, i.e. the backbone amine has been removed. 'K^{xxx}' gives the placement of the Lys residue on which acylation occurs. If no 'K' is given, the acylation is on Lys26*. The acylated side chains R₁, R₂, and R₃ are attached to Lys26*, Lys34*, and Lys38, respectively. The potency of each analogue-GLP-1R complex was taken from the study by Knudsen et al. (2000).

ID	R ₁	R ₂	R ₃	Mutation	His7* des-amino	EC ₅₀ (pM)
GLP-1 lira		—	—	— K34R*	✗ ✗	55 ± 19 61.0 ± 7.1
C16L,K ^{26,34}			—	—	✗	16700 ± 3700
C14A		—	—	K34R*	✗	72.0 ± 0.7
daC16L,K ³⁴	—		—	K26R*	✓	2360 ± 370
daC8,K ³⁴	—		—	K26R*	✓	236 ± 66
C12A,K ³⁸	—	—		K26R*, K34R*	✗	4.19 ± 0.98
C14L,K ³⁸	—	—		K26R*, K34R*	✗	54 ± 1

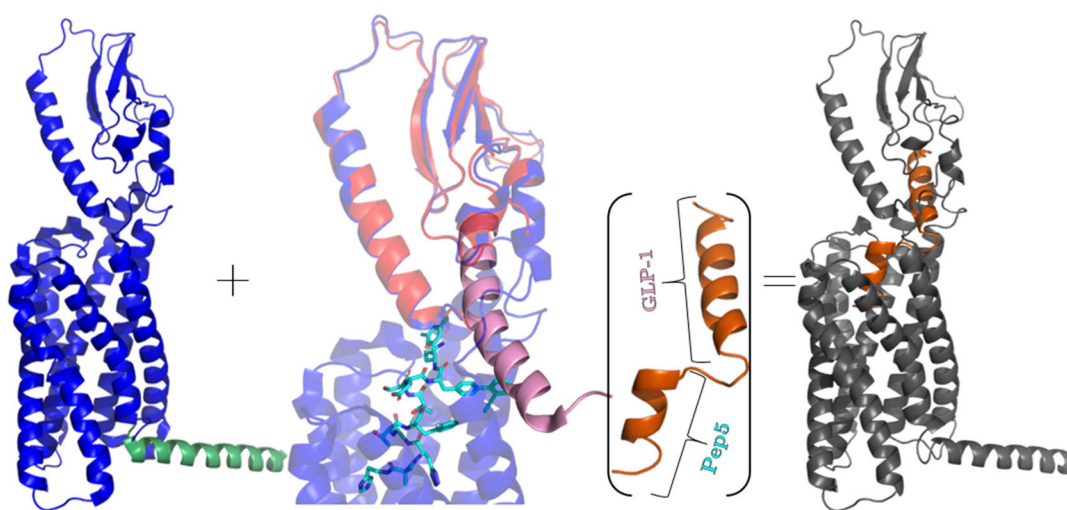


Figure 2. Receptor model creation based on the 5NX2 structure (blue ribbon) and the 4L6R structure (green ribbon). Position and model of GLP-1 (orange ribbon) based on GLP-1 (pink ribbon) in complex to the ECD from 3IOL (red ribbon) and Pep5 (cyan sticks) in 5NX2 (blue ribbon) in the middle. Resulting model structure of GLP-1 (orange ribbon) in complex with GLP-1R (grey ribbon) to the right.

(Gómez Santiago et al., 2018; Xiao et al., 2020), and receptor ECD conformation (Zhang et al., 2019). The aim of our study was to give a molecular understanding of the experimentally observed differences in potency (Knudsen et al., 2000) and to reveal key interactions. This was done by studying proximity interactions and binding energies extracted from MD

simulations of eight different receptor-peptide complexes as further discussed in the Materials and Methods section. using the molecular mechanics-Poisson-Boltzmann surface area (MM-PBSA) approach. The simulations reveal that fatty acid structure and position of acylation affect the binding of the peptide to the receptor.

Table 2. Interactions between GLP-1 and GLP-1R as reported by Jazayeri et al. (Jazayeri et al., 2017) (J), Santiago et al. (Gómez Santiago et al., 2018) (S), Underwood et al. (Underwood et al., 2010) (U), and Zhang et al. (Y. Zhang et al., 2017) (Z), the reference(s) of the reported interaction is given in “Source” column. † indicates residue cluster formation rather than individual interaction pairs. ECD hydro indicates the hydrophobic patch on the ECD consisting of residues Leu32, Thr35, Val36, Trp39, Tyr69, Tyr88, Leu89, Pro90, Trp91. ECLx means extracellular loop x. TMx refers to transmembrane helix x. E-TM is the ECD to TMD linker. α is the helix in the ECD N-terminus. Potency reduction from mutagenesis studies is given as reviewed by de Graaf et al. (Graaf et al., 2016). For a given receptor residue, only the mutation causing the highest potency reduction is shown. For comparison, the strength of the interactions between the receptor–peptide residues extracted from the simulation is also given (- indicates that the given interaction is not present in the simulations) as the average of the four replicates. The strength is based on the averaged percentage a given interaction occurs in the system trajectory: 25 % to ~30 % is very weak to weak, ~50–75 % is medium, and 75 % or above is strong.

GLP-1—GLP-1R interaction	Source	Potency reduction (x-fold)	Strength in simulations
His7*—Arg299 (ECL2)	Z	85 (Arg299Ala)	—
His7*—Trp306 (TM5)	Z	206 (Trp306Ala)	—
His7*—Glu364 (TM6)	S	15 (Glu364Ala)	Very weak
His7*—Glu387 (TM7)	J, S	10 (Glu387Asp)	Strong
Ala8*—Leu384 (TM7)	J	41 (Leu384Ala)	Medium
Ala8*—Leu388 (TM7)	J	208 (Leu388Ala)	Medium
Glu9*—Tyr145 (TM1)	Z	<5 (Tyr145Ala)	—
Glu9*—Tyr148 (TM1)	S	26 (Tyr148Ala)	Strong
Glu9*—Tyr152 (TM1)	J, S	<5 (Tyr152 His)	Strong
Glu9*—Arg190 (TM2)	J, S, Z	270 (Arg190Ala)	Strong
Glu9*—Leu388 (TM7)	S, Z	208 (Leu388Ala)	Weak
Glu9*—Ser392 (TM7)	S, Z	<5 (Ser392Ala)	—
Gly10*—Trp306 (TM5)	J	109 (Trp306Ala)	—
Thr11*—Arg299 (ECL2)	S, Z	85 (Arg299Ala)	—
Thr11*—Asp372 (ECL3)	J	59 (Asp372Ala)	—
Thr11*—Leu384 (TM7)	J	41 (Leu384Ala)	Weak
Phe12*—Leu141 (TM1)	J	<5 (Leu141Ala)	Strong
Phe12*—Leu142 (TM1)	J	Not measured	—
Phe12*—Tyr145 (TM1)	Z	<5 (Tyr145Ala)	Medium
Phe12*—Tyr148 (TM1)	J	26 (Tyr148Ala)	Strong
Phe12*—Leu384 (TM7)	J	41 (Leu384Ala)	Weak
Phe12*—Phe385 (TM7)	Z	<5 (Phe385Ala)	—
Phe12*—Leu388 (TM7)	J	208 (Leu388Ala)	Weak
Thr13*—Lys197 (TM2)	J, Z	630 (Lys197Ala)	Strong
Thr13*—Leu201 (TM2)	J	Not determined	—
Thr13*—Phe230 (TM3)	J	Not measured	Weak
Thr13*—Met233 (TM3)	J	70 (Met233Ala)	—
Thr13*—Trp297 (ECL2)	S	316 (Trp297Ala)	—
Thr13*—Thr298 (ECL2)	J	<5 (Thr298Ala)	—
Ser14*—Asn300 (ECL2)	J	501 (Asn300Ala)	—
†Ser14*—Trp297 (ECL2)	Z	316 (Trp297Ala)	—
†Ser17*—Thr298 (ECL2)	Z	<5 (Thr298Ala)	—
†Ser18*—Arg299 (ECL2)	J, Z	85 (Arg299Ala)	—
Asp15*—Arg380 (TM7)	J, S	1853 (Arg380Asp)	Very weak
Val16*—Pro137 (E-TM)	J	Not measured	—
Val16*—Leu141 (TM1)	J	<5 (Leu141Ala)	Medium
Val16*—Leu142 (TM2)	J	Not measured	—
Val16*—Tyr145 (TM1)	J	<5 (Tyr145Ala)	—
Val16*—Leu201 (TM2)	J	Not determined	—
Val16*—Lys202 (TM2)	J	Not measured	—
Val16*—Tyr205 (TM2)	J	62 (Tyr205Ala)	—
Ser17*—Ser31 (ECD)	J	Not measured	Weak
Ser17*—Leu32 (α)	J	<5 (Leu32Ala)	Medium
Ser17*—Trp33 (α)	J	Not measured	—
Ser17*—Met204 (TM2)	J	334 (Met204Ala)	—
Ser17*—Tyr205 (ECL1)	J	62 (Tyr205Ala)	—
Ser17*—Leu217 (ECL1)	J	<5 (Leu217Ala)	—
Ser17*—Gln221 (ECL1)	J	Not measured	—
Ser17*—Gly295 (ECL2)	J	<5 (Gly295Ala)	—
Ser17*—Thr298 (ECL2)	J	<5 (Thr298Ala)	—
Leu20*—Leu201 (TM2)	S, Z	Not determined	—
Leu20*—Met204 (TM2)	S, Z	334 (Met204Ala)	—
Glu21*—Arg299 (ECL2)	S	85 (Arg299Ala)	—
Lys26*—Glu128 (ECD)	U	<5 (Glu128Ala)	—
†Phe28*—ECD hydro	U	Not applicable	Strong
†Ile29*—ECD hydro	U	Not applicable	Medium
†Leu32*—ECD hydro	U	Not applicable	Strong
Trp31*—Gln211 (ECL1)	S, Z	<5 (Gln211Asp)	—
Trp31*—His212 (ECL1)	S, Z	<5 (His212Ala)	—
Val33*—Arg121 (ECD)	U	<5 (Arg121Ala)	Weak

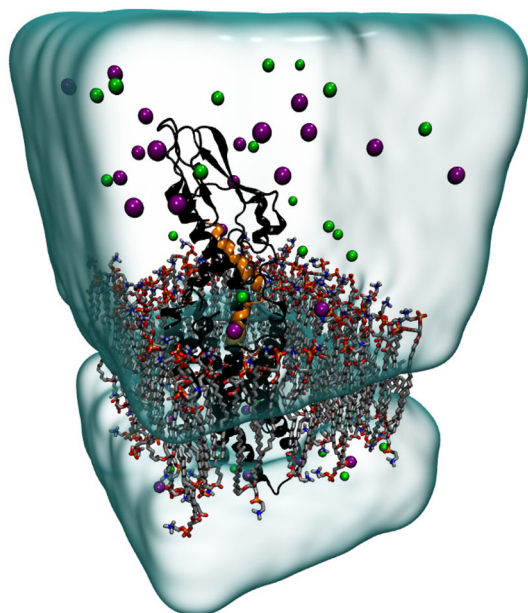


Figure 3. The GLP-1 system was inserted into a membrane bilayer patch, and solvated with TIP3 waters. The ionic strength was set by adding Na^+/Cl^- ions. Receptor and GLP-1 structures are shown in dark grey and orange ribbons, respectively. The membrane bilayer is shown in sticks with C-atoms coloured in light grey, phosphate coloured in orange, oxygen coloured in red, and nitrogen coloured in blue. The water solvent layers are shown as transparent surfaces. Na^+ and Cl^- ions are shown as purple and green spheres, respectively.

2. Materials and methods

2.1. GLP-1 receptor

The simulations were started from a full-length structure of GLP-1R that was built from the crystal structure of the human GLP-1 receptor in a complex with a truncated peptide agonist (PDB ID 5NX2 (Jazayeri et al., 2017)) and a class B GPCR (PDB ID 4L6R (Siu et al., 2013)) that shows 39% identity to GLP-1R. 5NX2 only contains the residues from 29 to 417, covering the ECD and the TMD, but not the intracellular C-terminal helix (residues 410 to 431). Hence, a complete receptor structure was created by making a model structure using residues 29-405 from 5NX2 (chain A) and the residues 406-431 from 4L6R (corresponding to 404-429 in the 4L6R structure). The model structure was constructed using the Structure Prediction Wizard in Maestro (Schrödinger, 2018), with the alignment of the GLP-1R sequence to the two template structures created using ClustalW (Larkin et al., 2007; Thompson et al., 1994) available through Maestro (Schrödinger, 2018), the alignment is shown in Table S1 in the supporting material (Materials and methods).

The structure of the resulting receptor model is shown in Figures 2 and S1 with structural elements indicated. The only solved structure of liraglutide is deposited in an 'L' shape structure (PDB ID 4APD), to build a liraglutide-receptor complex, the peptide agonist was designed in the L shaped format and the TMD and ECD were built separately. GLP-1R complexes have been studied using cryo-EM (Deganutti et al., 2022; Zhang et al., 2017, 2021; Zhao et al., 2022), and the liraglutide like peptide complex model can be built using many different combinations. Santiago et al. used a combination of 5VAI and 5NX2 to build activate GLP-1R model

(Gómez Santiago et al., 2018), 3IOL and 5NX2 were chosen instead of a rabbit GLP-1R structure, 5VAI.

2.2. GLP-1R–GLP-1 complex

The placement of GLP-1 in the receptor was based on the 5NX2 ligand and the C-terminal part of the GLP-1 ligand in the PDB ID 3IOL (Underwood et al., 2010) structure. The former is a ten residue peptide that resembles the N-terminal part of GLP-1, called Pep5, situated in the top part of the TMD of the receptor in 5NX2. The latter is a crystal structure of only the GLP-1R ECD in complex with GLP-1(7*-35*) in a straight 'I' shape.

To begin with, the residues in Pep5 were changed to the corresponding GLP-1(7*-17*) residues (see Table S2).

Hereafter, the ECD from the 3IOL structure was aligned to the ECD part in the 5NX2 structure to ensure the right orientation of the ECD towards the TMD. This is illustrated in Figure 2.

Following this, the structure of GLP-1, with the correct orientation with regards to both the ECD and TMD, was obtained by creating a homology model. This homology model of GLP-1 residues 7*-35* was constructed (same procedure as described for the receptor) using residues 7*-17* from the mutated Pep5 in 5NX2 and residues 18*-35* from GLP-1 in 3IOL as templates. – as shown in Table S3.

The new GLP-1(7*-35*) structure was elongated to GLP-1(7*-36*) by adding an Arg residue to the C-terminus, using Maestro. Hereafter, residues 16*-21*, around the bend, were minimised. Finally, residues 32*-36*, in the C-terminal part, were minimised. Both minimisation steps were carried out using the built-in 'Cleanup' tool in Maestro (Schrödinger, 2018).

2.3. GLP-1 analogues

To create the remaining GLP-1R–peptide complexes, the GLP-1 structure (Figure 2, GLP-1) was modified manually in Maestro to give the seven other peptide structures. The analogue structures were based on the results of an extensive screening of different acylation strategies, where the C-terminal part of GLP-1 was acylated to screen for potent GLP-1 analogues (Knudsen et al., 2000). The structures as well as their respective IDs are given in Table 1 and Figure S4. Liraglutide was created by aligning Lys26* in GLP-1 to the acylated Lys26* in the NMR structure of liraglutide (PDB ID 4APD) and adding the L-γ-Glu linker and C16FA chain atoms to the GLP-1 structure. Then Lys34* in GLP-1 was mutated into Arg34*, and Gly37* was added to the C-terminal helix chain (Figure S2, lira). C16L,K^{26,34} was produced by altering the liraglutide structure. An extra L-γ-Glu linker with an attached 16C-atom FA chain was added to Lys34* (Figure S2, C16L,K^{26,34}). C12A,K³⁸ was constructed based on the GLP-1 structure where both Lys26* and Lys34* were mutated into Arg. Furthermore, the C-terminus was elongated with Arg37* and Lys38*. The FA chain atoms from liraglutide were added to Lys38*, and afterwards shortened to 12C-atoms with an added acid group in the ω tail of the FA chain

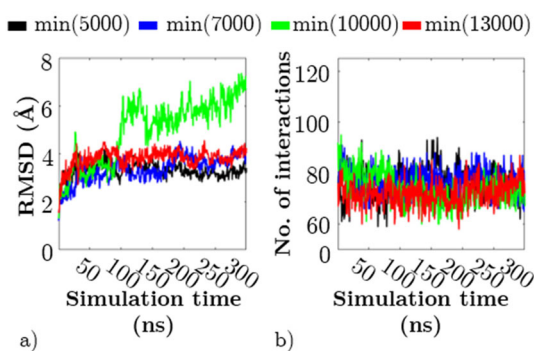


Figure 4. a) The total RMSD for the GLP-1 system, calculated for the receptor plus the peptide backbone atoms. b) The total number of interactions between the receptor and GLP-1 are shown as a function of simulation time. An interaction is counted when a residue in the receptor is within 3 Å of a residue in the peptide. The different colours indicate the individual replicate simulations differing only in the number of initial minimisation steps, i.e. 5000 (black), 7000 (blue), 10000 (green), or 13000 (red) steps.

(Figure S2, C12A,K³⁸). C14A was constructed from liraglutide, by removing the L-γ-Glu linker and shortening the FA chain to only 14 C-atoms, and adding a free acid group in the FA chain end (Figure S2, C14A). daC8,K³⁴ was made from GLP-1, by mutating Lys26* to Arg and adding the FA chain from liraglutide to Lys34*, which was shortened to eight C-atoms. Also, the C-terminus was elongated with Arg37*, and the amino group on the His7* backbone was replaced by a hydrogen atom (Figure S2, daC8,K³⁴). daC16L,K³⁴ was created from daC8,K³⁴ by replacing the FA chain with the L-γ-Glu linker and 16 C-atom FA chain from liraglutide (Figure S2, daC16L,K³⁴). C14L,K³⁸ was modelled from C12A,K³⁸ by exchanging the FA chain on Lys38* to the L-γ-Glu and FA chain from liraglutide, shortened to a 14 C-atom FA chain (Figure S2, C14L,K³⁸). The structures are listed in Table 1, together with their potencies toward GLP-1R as given by Knudsen et al. (2000).

The result is eight different receptor–peptide complexes. psfgen (Gullingsrud et al., 2017), a VMD (Humphrey et al., 1996) plugin, was used to create PDB and structure files for the systems. The four disulphide bonds in the receptor (Cys46-Cys71, Cys62-Cys104, Cys226-Cys296, and Cys126-Cys85) were defined as patches. Then the protein complexes were inserted in a membrane patch and solvated as described in the supporting material (Materials and methods). An example of a system is shown in Figure 3.

The size of the system box is (83.5 × 80.3 × 121.0) Å³ for all systems, ensuring at least 15 Å between each protein molecule when using periodic boundary conditions. Simulations were checked and showed that there were no contacts between mirror protein molecules throughout the simulation time, ensuring that the system box sizes were sufficient. One Cl⁻ atom was added to neutralise the systems, then 26 NaCl molecules were added to each system to give an ionic strength of 100 mM. The number of water molecules varies from 13628 to 13685 giving a total atom count varying from 62886 to 63306. For details, see Table S4. Each system was run for 300 ns in four replicates amounting to a total simulation time of 1.2 μs for each complex.

2.4. Simulation setup

The systems were minimised and modelled according to the AMBER tutorial on MD simulations of a protein in a lipid bilayer (Madej & Walker, 2014). All simulations were run in AMBER (Case et al., 2016) using the CHARMM36 force field (MacKerell et al., 1998). Additional CHARMM36 topologies/parameters were given to accommodate the acylation of Lys – values are shown in the supporting material under ‘Materials and methods’.

Each system was simulated in quadruplets differing only in the number of initial minimisation steps. Hence, four sets of eight GLP-1R–peptide complexes, totalling 32 simulations. Each system was minimised for either 5000, 7000, 10000, or 13000 steps to obtain varying starting conditions. The individual systems are referred to as the complex ID followed by the minimisation step number in parenthesis, meaning that the simulation of the GLP-1R–GLP-1 complex started from the minimisation protocol of 7000 minimisation steps, will be written as GLP-1R–GLP-1(7000).

Minimisation was performed using the steepest descent algorithm for the first 5000 steps and the conjugate gradient algorithm for the remaining steps. Non-bonded energies were cut-off at 12 Å with a switching function from 10 to 12 Å to accommodate the use of the CHARMM36 force field.

After the minimisation, the systems were subjected to a heating ramp in two steps. First, the lipids were held constant, while the systems were heated at a constant volume from 0 K to 100 K using a Langevin thermostat (Loncharich et al., 1992) with a collision frequency of 1 ps⁻¹. The second heating went from 100 K to 303 K, but this time under constant pressure at 1 bar, controlled anisotropically (since the system contains a membrane), using the Berendsen barostat (Berendsen et al., 1984) with a pressure relaxation time of 2 ps. Finally, the production was performed in an *NPT* (constant number of particles (*N*), pressure (*P*), and temperature (*T*)) ensemble, at 303 K and 1 bar, controlled using a Langevin thermostat and a Monte Carlo barostat respectively (Åqvist et al., 2004; Chow & Ferguson, 1995; Loncharich et al., 1992). The SHAKE algorithm (Gonnet, 2007; Miyamoto & Kollman, 1992; Ryckaert et al., 1977) was used to restrain hydrogen bonds. The non-bonded cut-off distance was set to 12 Å with a switching function for the Lennard-Jones potential from 10 Å to 12 Å. The particle-mesh-Ewald (PME) method (Darden et al., 1993; Essmann et al., 1995) was used for the evaluation of long-range electrostatics. The production run was continued for 300 ns.

2.5. MM-PBSA binding free energies

In order to rank the GLP-1 analogues, we have applied the computationally efficient MM-PBSA method (Equation (1)) to compute binding free energy estimates, Δ*G*_{bind}. A more rigorous approach to estimating the binding energy would be the application of more advanced free energy methods such as metadynamics, which was applied by Timothy Clark and co-workers for a related GPCR member where crystal structures of the two end-point states are known (Saleh et al., 2017a, 2017b, 2017c). Since there is no crystal structure of the open state

available for GLP-1R, we focused on the complex conformation and applied MM-PBSA to estimate the binding energy. The computational efficiency of the MM-PBSA method is achieved by introducing several approximations that have been debated in the literature (Genheden & Ryde, 2015; Gohlke & Case, 2004; Kollman et al., 2000; Liang et al., 2009; Srinivasan et al., 1998), since the method relies on using a constant value for the dielectric constant and a simplified method to estimate conformational entropy. Although several approximations are introduced, the method has been applied successfully in many aspects of drug design (Wang et al., 2019), including e.g. ranking of ligands during a structure-based screening of protein-protein complexes (Ryberg et al., 2020). The MM-PBSA energy is computed as follows:

$$\Delta G_{\text{bind}} = G_{\text{MMPBSA, com}} - G_{\text{MMPBSA, rec}} - G_{\text{MMPBSA, pep}} \quad (1)$$

Where the MM-PBSA energies for the individual components are calculated from one simulation of the complex according to:

$$G_{\text{MMPBSA, xxx}} = E_{\text{int, xxx}} + G_{\text{solv, xxx}} - TS_{\text{xxx}} \quad (2)$$

Internal energies ($E_{\text{int, xxx}}$), which is the sum of the bond, angle, dihedral, van der Waals, and electrostatic energies were calculated using the VMD (Humphrey et al., 1996) plugin, NAMDenergy (Nelson et al., 1996), using the CHARMM36 force field (MacKerell et al., 1998) and the PME method (Darden et al., 1993; Essmann et al., 1995). The solvation energy ($G_{\text{solv, xxx}}$) is a sum of the non-polar and the polar solvation energies. Non-polar solvation energies were calculated from the solvent-accessible surface area (SASA), applying the linear relation:

$$G_{\text{solv, non-polar}} = \gamma \text{SASA} + \beta \quad (3)$$

With the surface tension, $\gamma = 0.00542 \text{ kcal}/(\text{mol } \text{\AA}^2)$ and $\beta = 0.92 \text{ kcal/mol}$ (Abroshan et al., 2010; Karplus & Kushick, 1981; Sitkoff et al., 1994). SASA was computed in VMD using a probe radius of $1.4 \text{ \AA}$ corresponding to the spherical size of a water molecule. The polar solvation energy was calculated using APBS (Adaptive Poisson-Boltzmann Solver) (Jurrus et al., 2018). Here, dielectric constants of 80 for water (solvent) and 1 for protein (solute) were used, together with a probe radius of $1.4 \text{ \AA}$. Noteworthy, studies have indicated that the resulting solvation energies depend highly on the used dielectric constant for the protein (Feig et al., 2004; Schutz & Warshel, 2001). PQR files were generated using PDB2PQR (Dolinsky et al., 2004, 2007).

All of these energetic contributions were calculated for 200 frames, 0.5 ns apart, covering the last 100 ns of the simulations.

Entropy calculations were investigated for convergence. The GLP-1R-C16L_{K26,34}(7000) simulation was chosen as a representative of all the systems and was simplified to only span the α carbons including the C atoms in the FA chains. Entropy values were calculated using the Schlitter approach (Schlitter, 1993):

$$S = \frac{1}{2} k_B \ln \det \left(1 + (k_B T e^2 / \hbar^2) \mathbf{M} \sigma_c \right) \quad (4)$$

where 1 is the identity matrix, $e = \exp(1) = \text{Euler's number}$, $\mathbf{M}$ is the mass matrix containing the masses on the diagonal,

and σ_c is the variance-covariance matrix of the Cartesian coordinates obtained directly from the MD simulation output. $\hbar$ is Planck's constant divided by 2π .

The resulting entropy values were calculated for an increasing number of frames. The result is plotted in Figure S3 (supporting material, 'Materials and methods' section).

The results indicate that the entropy starts to converge when using 2500 frames in each of the two batches. Hence, all entropy values were calculated by use of two batches with 2500 frames in each batch. The frames were evenly spread out over the last 100 ns with a step size of 20 ps. The variance-covariance matrices were calculated directly from the trajectory files using ProDy (Bakan et al., 2010; Bakan et al., 2011) and the entropy values were then calculated using Equation (4).

3. Results and discussion

In the following, we first present our results for model validation, followed by a discussion on the point of equilibration of the different systems. Lastly, we address the physicochemical interactions between the extracellular domain (ECD) and the FA chains to shed light on the structure-activity relationship responsible for the observed experimental differences in potency, which might explain the different biological responses.

3.1. Simulation model validation

The described model setup was validated by investigating the simulations of the GLP-1R—GLP-1 system.

3.1.1. Overall stability

A good estimate of the overall stability of the system can be found by taking a look at the root mean square deviation (RMSD) for the whole system (receptor plus peptide backbone atoms) and the total number of proximity interactions between the receptor and the peptide. Both are monitored for 300 ns with a time interval of 0.5 ns.

Inspection of the RMSD plotted in (Figure 4a) shows overall convergence for three out of four replicates with a maximum deviation of about $4 \text{ \AA}$. The simulation replicate minimised for 10000 steps (GLP-1R—GLP-1(10000)) increases to $\sim 7 \text{ \AA}$. Further inspection of the RMSD for the individual domains (see Figure S4) showed that there is less than $1 \text{ \AA}$ displacement within the ECD and the transmembrane domain (TMD), respectively, and that the increase in total RMSD stems from a rotational displacement of the ECD performed at around 100 ns resulting in a $\sim 2\text{--}3 \text{ \AA}$ change in RMSD. To investigate the effect of the rotation on GLP-1R—peptide interactions, we calculated the time evolution of the number of proximity interactions (Figure 4b). The number was updated when the distance between two residues, one in the receptor and one in the peptide, was within $3 \text{ \AA}$ of each other.

For the simulation GLP-1R—GLP-1(10000) and the other three replicates, the total number of proximity interactions is

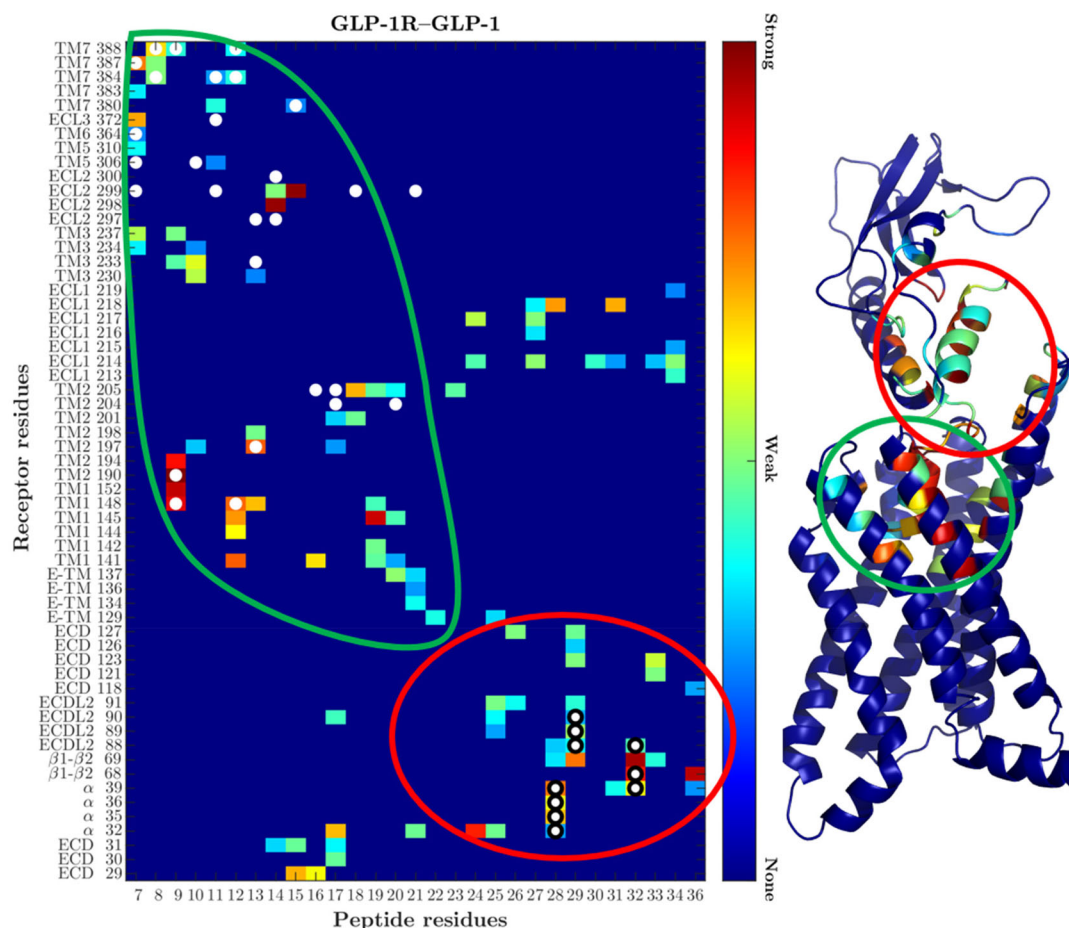


Figure 5. To the left: Heatmap of the receptor–peptide interaction pattern. The peptide and receptor residues are on the x and y axes, respectively, and interaction pairs are given by the coloured squares. The colour illustrates the binding strength of the individual pairs according to the colour bar. The dots represent interactions given in Table 2: White are those where the measured potency reduction is greater than 5-fold, and white with black edge are the ones reported by Underwood et al., 2010. To the right: The structure of the complex with residues coloured according to the interaction strength in the heatmap. The interactions between the GLP-1 C-terminal end and the receptor ECD are indicated in red circumferences. The interactions between the peptide N-terminal end and the top part of the TMD in the receptor are indicated in green circumferences.

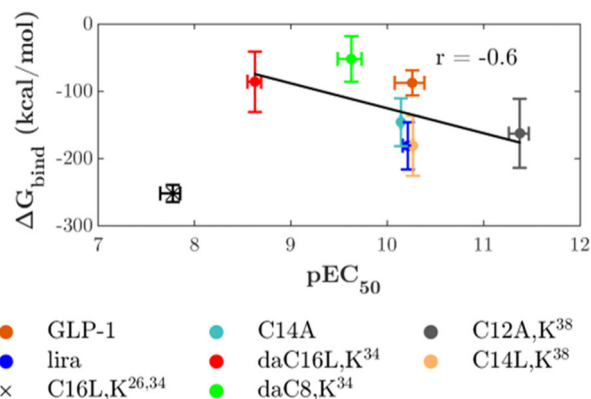


Figure 6. The calculated binding energies plotted against the potency values (shown as $pEC_{50} = -\log(EC_{50})$). The C16L,K^{26,34} system (black x) is not included in the computed regression line and correlation coefficient, r , shown in the plot.

stable throughout the 300 ns simulations. There is no apparent correlation between the change in RMSD and the number of proximity interactions showing that the rotation of the extracellular domain observed in the simulation GLP-1R–GLP-1(10000) has no effect on the GLP-1R–peptide interactions.

3.1.2. Important GLP-1R–GLP-1 interactions

Specific interactions between GLP-1R and GLP-1, based on previous studies, are compared to those of the GLP-1R–GLP-1 simulations to further evaluate the validity of the simulations. Such studies cover both potency studies as well as mutagenesis studies of GLP-1 and GLP-1R (Adelhorst et al., 1994; Graaf et al., 2016). Runge et al. (Runge et al., 2008) crystallised the receptor ECD in a complex with Exendin-4 (PDB ID 3C59), and similarly, Underwood and co-workers (Underwood et al., 2010) could solve the structure of the receptor ECD in complex with GLP-1 (PDB ID 3IOL). Jazayeri et al. (Jazayeri et al., 2017) reported the full-length GLP-1R structure in a complex with a truncated 10 residue peptide agonist (PDB ID 5NX2), whereas Zhang and co-workers (Zhang et al., 2017) showed *Oryctolagus cuniculus* GLP-1R in complex with GLP-1 (PDB ID 5VAI). An overview of GLP-1R–peptide interaction studies is given in Table 2.

The important interactions between GLP-1 and GLP-1R are mainly located in two different areas. One area is comprised of the hydrophobic face (residues Phe28*, Ile29*, and Leu32*) in the C-terminal end of GLP-1, which interacts with the hydrophobic patch on the ECD (Adelhorst et al., 1994; Jazayeri et al., 2017; Runge et al., 2008; Underwood et al.,

2010). The other area is made up of the N-terminal part of GLP-1 and the top of the TMD structure, including the three extracellular loops (ECL1-3) (Adelhorst et al., 1994; Gómez Santiago et al., 2018; Jazayeri et al., 2017).

To shed some light on the binding interactions between GLP-1R and GLP-1 in the simulations, residue pair interactions (GLP-1R and GLP-1 residues within 3 Å of each other) were monitored as a function of simulation time, i.e. frames. Since there are four replicas, there will also be four sets of interacting pairs with associated frame counts. Hence, these four frame counts were averaged to give one total frame count per identified interaction pair. Peptide residues are plotted on the x-axis and receptor residues on the y-axis. The interactions are then plotted as specific pairs in a heatmap, where the colourscales disclose the strength of the specific interaction. The higher the frame count for a specific interaction pair, the stronger is the interaction. Interactions occurring in less than 25% of the total frames are not shown and are interpreted as negligible. A frame count corresponding to 25% to 75% is considered as a weak to medium interaction, and 75% or above, as a strong interaction. The results are shown in Figure 5.

From the heatmap in Figure 5, it is evident that our simulation captures several important interactions between His7*, Glu9*, Phe12*, Thr13*, and Asp15* and the receptor transmembrane part. Gly10* does not show interaction with receptor residues, suggesting that Gly10* ensures the overall flexibility of the peptide chain when binding to the activated receptor. The heatmap also shows that our simulation system can obtain and maintain the important interactions between the hydrophobic patch noted in the literature (Jazayeri et al., 2017; Runge et al., 2008; Underwood et al., 2010) and GLP-1. Specifically, the patch (comprised of the ECD β -strand 1 to β -strand 2 turn (β 1- β 2) (Glu68 and Tyr69), α -helix (Leu32, Thr35, Val36, and Trp39), and ECD loop 2 (ECDL2) (Tyr88, Leu89, Pro90, Trp91)) interacts with the hydrophobic residues Phe28*, Ile29*, and Leu32* in the C-terminal part of GLP-1. This is shown in a close-up of the ECD and peptide C-terminal part in Figure S7.

In conclusion, our GLP-1R–GLP-1 simulations can capture several important interactions reported previously, in structural as well as mutagenesis studies.

3.1.3. Binding free energy and potency

Binding free energies for all the receptor–peptide systems are compared to experimentally measured potency values to consider a more overall trend amongst the simulations. Therefore, the binding free energies, ΔG_{bind} , are calculated using the MM-PBSA approach for the eight receptor–peptide systems. These are compared to potency values for the peptides reported by Knudsen et al. (2000). In Figure 6, the calculated binding free energies and the potency values (in the form of $\text{pEC}_{50} = -\log(\text{EC}_{50})$) are plotted against each other (Acevedo et al., 2018; Chen et al., 2014; Tang et al., 2018; Zhang et al., 2011). The MM-PBSA energies are calculated as an average from the individual energies calculated for each of the four simulation replicas. MM-PBSA energies are only

calculated for the last 100 ns of the simulations that cover the converged part of the trajectories.

The C16L,K^{26,34} system is regarded as an outlier for several reasons. First of all, it is a structural outlier because it contains two acylated FA chains, whereas the others only have one. Secondly, the two FA chains self-interacting in multiple ways during the simulations. We see a correlation between potency and binding energies for single FA chain peptides, in contrast to C16L,K^{26,34} with two FA chains. We hypothesise that the self-interacting FA chains will prevent the initial binding of the peptide C-terminus to the ECD, and therefore hinder the entry of the peptide into the receptor (see ECD—peptide interactions for further detail). Lastly, the measured potency value can be considered an outlier in both the reduced dataset used here, and in the full dataset reported by Knudsen, (2010) by statistical analyses (data not shown). Therefore, the C16L,K^{26,34} point is not considered when calculating the correlation coefficient, which is then determined to be -0.6 . This indicates that there exists a moderate positive correlation between potency values (EC_{50}) and the calculated MM-PBSA energies.

3.2. Equilibration of systems

As for GLP-1, first, we consider the RMSD and total proximity interaction counts for all systems, to ensure that all the simulations are converged. The results are shown in Figures S5 and S6.

From the RMSD (Figure S5), we see that the majority of the systems have overall converged. For a few systems, individual replicates – besides GLP-1R–GLP-1(10000), these are GLP-1R–C14A(13000) and GLP-1R–C12A,K³⁸(13000) – do not converge with respect to RMSD. As discussed previously, this is due to the slight rotation of the ECD during the simulation (see Figure S4), and as shown by the total number of proximity interactions, the rotation has no effect on the GLP-1R–peptide interactions (Figure S6).

The number of proximity interactions, overall, converges after 100–200 ns for all systems, and fluctuates with approximately ± 10 around the average value of proximity interactions. Although the RMSD of a few replica simulations is not fully converged, the number of proximity interactions is stable, indicating that the GLP-1R–peptide interactions are maintained throughout the simulations of all replicates of a system and that the simulations of the eight different GLP-1R–peptide complexes are converged.

Adding this to the previous findings for the GLP-1 system and concerning the potency- ΔG_{bind} correlation, we conclude that the created model structure and the performed simulations of the eight different receptor–peptide complexes are valid to use for further analyses.

3.3. ECD—peptide interactions

Looking at the interaction patterns, it is evident that the main differences in binding stem from the FA chain interactions with the receptor, as well as the hydrophobic interactions between the patch on the ECD and the peptide

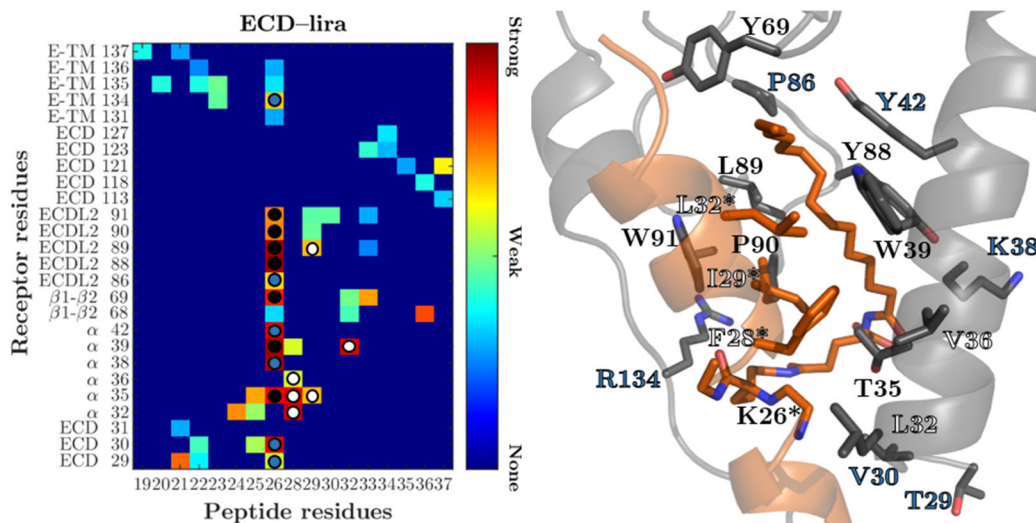


Figure 7. Heatmap showing the interactions between the receptor ECD and liraglutide, left. This is shown as the sum over all four replicas. Important interactions are noted with a dot. Black dots denote interactions between the ECD hydrophobic patch and peptide FA chain. White dots with a black edge show interactions between the ECD hydrophobic patch and peptide residues Phe28*, Ile29*, and Leu32*. Blue dots with a black edge represent interactions between the ECD and the FA. The ECD residues (grey sticks) interacting with the acylated Lys26* residue in liraglutide as well as the hydrophobic residues Phe28*, Ile29*, and Leu32* (orange sticks), right. Residue labels are coloured according to the dots in the heatmap on the left. The image is taken from the last frame of the GLP-1R-lira(7000) simulation.

hydrophobic face, i.e., residues Phe28*, Ile29*, and Leu32* (see Figure S8–S14 for overall interaction patterns). This is with the exception of the two de-amino peptides (daC16L,K³⁴ and daC8,K³⁴), where another main difference is that His7* has no strong interactions with Glu387 (see Figures S11 and S12, respectively). Therefore, we have chosen to discuss the FA chain orientation, the proximity ECD–FA interactions, as well as the proximity interactions between the ECD patch and the hydrophobic face on the peptide C-terminus in the following section. For simplicity, a ‘close-up’ of the interaction pattern, covering only the ECD and peptide C-terminal part is given together with the corresponding part of the structure. As an example, the GLP-1R—lira complex is discussed here, whereas the remaining systems, i.e., GLP-1R–C16L,K^{26,34}; GLP-1R–C14A; GLP-1R–daC16L,K³⁴; GLP-1R–daC8,K³⁴; GLP-1R–C12A,K³⁸; and GLP-1R–C14L,K³⁸ are discussed in the supporting material (see Figure S15–S20).

The binding interface surrounding the hydrophobic patch on the ECD and the FA chain on liraglutide is shown in Figure 7.

The FA chain in liraglutide forms several hydrophobic interactions with the hydrophobic patch made up of the ECD α -helix, the β 1– β 2, and ECDL2 (Figure 7, left), which means that the FA chain on liraglutide can neatly tug into the hydrophobic pocket on the ECD, without preventing interactions between the ECD patch and residues Phe28*, Ile29*, and Leu32* (Figure 7, right). Nearly all of these interactions are strong, and generally, only a few weak interactions are present in the FA chain.

Similar detailed analyses are given in the supporting information, but the major differences in ECD—peptide binding between the eight receptor—peptide complexes are summed up in Figure 8 together with the calculated MM-PBSA binding energies.

A closer look at the individual interactions between the peptide C-termini and receptor ECD in the simulated systems and the binding free energies suggest that the position of acylation is of great importance. If the FA chain is attached to Lys34*, it points away from the ECD α -helix, and the FA chain is thus hampered in forming hydrophobic interactions with the ECD, resulting in the lower binding affinities for daC16L,K³⁴ (–86 kcal/mol) and daC8,K³⁴ (–52 kcal/mol). This finding correlates well with the lower potencies reported by Knudsen et al. (2000) for species with acylation on Lys34*. If the acylation is on Lys26*, the FA chain, on the other hand, points towards the ECD α -helix, and can snuggle up against the hydrophobic patch on the ECD, resulting in more negative binding energy for liraglutide (–181 kcal/mol), and higher potencies for peptide species with acylation on Lys26* (Knudsen et al., 2000). If acylation occurs on Lys38*, the added flexibility of an additional residue makes it possible for the peptide backbone to bend, allowing the FA chain to interact with the apolar patch, which correlates with the higher binding affinities for C12A,K³⁸ (–163 kcal/mol) and C14L,K³⁸ (–181 kcal/mol). This correlates with the higher potencies for species with acylation site on Lys38* (Knudsen et al., 2000). In addition to this, the results indicate that acylation on both Lys26* and Lys34* causes bulky self-interaction between the two FA chains. However, C16L,K^{26,34} sustains high binding energy of –252 kcal/mol, due to an increased number of interactions between the two self-interacting FA chains and another hydrophobic part of the ECD, located away from the hydrophobic patch involved in GLP-1 binding. We believe that the bulkiness of the FA chains will prevent insertion into the receptor, and hence receptor activation, explaining why the potency is lower for this species (Madsen et al., 2007). This could be studied further by e.g. performing metadynamics simulations of the systems to explore the entry pathway of the peptides into the receptor (Darve et al., 2008; Saleh et al., 2016). Similar results were

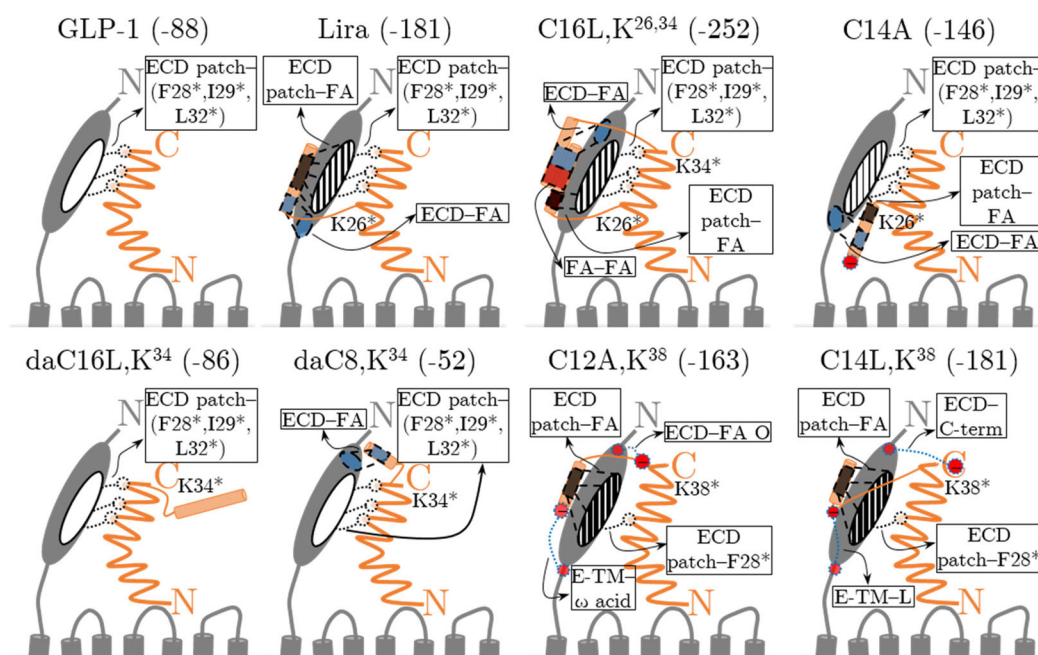


Figure 8. Schematic illustration of the differences in ECD—peptide C-terminus binding. The receptor is shown in dark grey with the ECD as an ellipse and the top of the TMD as cylinders inside the membrane (lighter grey band). The peptides are shown in orange with FA chains as cylinders attached to the peptide helix. Interactions between the hydrophobic patch on the ECD (large ellipse on ECD) and the hydrophobic face on the peptide C-terminus comprised of the residues Phe28*, Ile29*, and Leu32* (white spheres) are represented via dotted lines between white areas. Hydrophobic interactions between part of the FA chain (black rectangle) and the ECD patch are shown as dashed lines between black areas. Interactions between the part of the FA chain (blue rectangle) and the part of the ECD (little blue ellipse) are indicated by dashed lines between blue areas. Electrostatic interactions are indicated by blue dotted lines between charges (red spheres with blue edges). The hydrophobic patch on the ECD is coloured according to the relative interaction type, that is, all white means no interaction to the FA chain, and black/white means interactions to both the FA and hydrophobic face on the peptide helix. The less black, the fewer the interactions to the FA chain, and vice versa, the less white, the fewer interactions to the peptide hydrophobic face. The calculated MM-PBSA binding energies (kcal/mol) are given in parenthesis next to the peptide IDs.

seen, in the form of lower potencies, for peptide species with bulky FA chains attached to Lys26* (Madsen et al., 2007). The length of the FA chain, including the extra length gained from a linker, also appears to be of importance, pointing to the necessity of a certain length to reach across the full span of the hydrophobic pocket on the ECD. For instance, the shorter FA chain on C14A provided relatively weaker interactions with ECD compared to liraglutide, even though C14A was acylated on Lys 26*. In addition, the acylated sidechain pointed downwards compared to the FA that is attached to liraglutide (Figures 8 and S16). This MD result can be compared to the cryo-EM structure of the semaglutide-GLP-1R complex (Zhang et al., 2021). The cryo-EM structure shows that the acylated sidechain on Lys26* can either interact with TM1 or the ECD. The latter interactions are also seen in our MD simulations. However, considering that semaglutide structure has two distinct FA configurations, it is evident that FA can also point away from ECD. In contrast, a cryo-EM study on tirzepatide, containing an acylated sidechain on Lys26*, bound to the GLP-1R (Zhao et al., 2022) revealed that no distinct conformations were observed for the acylated sidechain. The result from literature (Zhang et al., 2021; Zhao et al., 2022) and our simulations indicate that the composition of FA or GLP-1 analogue can affect the flexibility of the FA sidechains. Furthermore, the presence of an ω acid group seems to influence binding to the ECD, dependent on where acylation is located. If a free acid group is present on Lys38* (as in C12A,K³⁸), polar interactions to

the ECD to TMD linker (E-TM) can stabilise the position of the FA chain along the ECD α -helix. Additionally, the presence of a charged linker appears to facilitate hydrogen bonding with Arg131, contributing to a stable placement of the FA, if the acylation site is on Lys38*, as in C14L,K³⁸. However, acylation on Lys38* with the FA chain stabilised by electrostatic interaction, causes the peptide helix to sit slightly further away from the ECD compared to GLP-1. This results in fewer interactions between the peptide hydrophobic face and the ECD patch. In summary, our MD simulations provide an overview of the interactions between a variety of GLP-1 analogues with GLP-1R and demonstrate that our methodology can be used in the design of the different types of GLP-1 analogues.

4. Conclusion

We have performed a series of MD simulations of different receptor–peptide complexes. MM-PBSA energies were calculated using selected frames from the MD trajectories, and the average binding energies were compared to experimentally determined potencies. The resulting correlation coefficient of -0.6 indicates a moderate positive correlation between potency and MM-PBSA energy for single acylated peptides.

Although the overall binding of the peptides to the receptor is similar, significant differences are seen in how the FA chains interact with the ECD. The fatty acid chain structure

as well as the position of acylation in the peptide affect peptide binding. Acylation on Lys26* results in more ECD–FA interactions than observed when acylation is on Lys34*, resulting in higher potency in the former case. Acylation on both Lys26* and Lys34* may cause bulky FA self-interactions that could hamper initial receptor binding, thus reducing efficacy. Acylation on Lys38* creates flexibility to the peptide backbone, making FA interactions with the ECD possible, leading to higher potencies.

In addition to this, it appears that the presence of a charged linker can potentially stabilise the position of the FA chain by forming hydrogen bonds to arginine residues in the E-TM linker region in the receptor. Furthermore, having a free acid group in the ω FA tail can seemingly be beneficial if the acylation occurs on Lys38*, where it can stabilise the position of the FA chain on the hydrophobic patch in the ECD.

Our findings in the structure-activity relationship of different acylated GLP-1 analogues might be used to guide the development of new acylated potent peptide drugs, especially concerning the positioning of the acylation site and FA chain bulkiness.

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Disclosure statement

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